Plasmid preparation by alkaline lysis

Under alkaline conditions, high molecular weight DNA is selectively denatured while covalently closed circular plasmid DNA remains double-stranded. When the solution is rapidly neutralized, denatured chromosomal DNA forms an insoluble clot, while plasmid DNA remains in solution (Birnboim and Doly, 1979).

This method yields 5–20 µg of highly pure plasmid DNA from mini-preps (a few milliliters of culture), and up to milligrams from a liter-scale culture.

The plasmids are purified over a silica matrix (*Alternative A*), by alcohol precipitation (*Alternative B*), or over an anion-exchange column (*Alternative C*).

Risk assessment

– Guanidine salts form TOXIC GASES with bleach or strong acids

▷ Wear gloves, safety glasses, lab coat

◻ DO NOT add bleach or acidic solutions directly into sample waste



Reviewed: Oct 26, 2026

Procedures

›› ***Alkaline lysis***

◻ LB medium (Miller) ⟋ R0030
◻ Buffer P1 (Resuspension buffer), pH 8.0 ⟋ R0006
◻ Buffer P2 (Lysis buffer) ⟋ R0007
◻ Buffer N3 (Neutralization buffer), pH 4.8 ⟋ R0009

(1.) Grow bacteria in 1–5 mL LB (Miller) overnight to stationary phase. Collect the cells by centrifugation.

> ***Hint:*** For culture tubes, spin at 3 000 × *g* for 5 min. For 2 mL centrifuge tubes, a brief spin at 11 000 × *g* is sufficient.

(2.) Resuspend the pellet in 150 µL Buffer P1.

> ***Hint:*** Buffers P1, P2, N3, PB, and PE refer to standard silica-column miniprep kit components (e.g., QIAGEN QIAprep). Equivalent buffers from other suppliers can be substituted.

(3.) Add 250 µL Buffer P2. Mix by gently inverting the tube 4–6 times. Incubate at room temperature for no more than 5 min or until the lysate appears clear.

> ***Critical:*** Do not vortex or pipette to avoid shearing chromosomal DNA.

(4.) Add 350 µL Buffer N3. Mix gently by inversion 4–6 times.

> ***Critical:*** Ensure complete neutralization to precipitate protein and chromosomal DNA. If Buffer P2 contains thymolphthalein, mix until the solution is colorless.

(5.) Centrifuge at 17 000 × *g* for 3 min to pellet precipitated material. Repeat if the lysate remains turbid.

(6.) Transfer the clear supernatant to a fresh tube for purification. Avoid any carryover of precipitate.

A › ***Purification of plasmid DNA over a silica matrix***

◻ Buffer PB (Binding buffer), pH 5.0 ⟋ R0010
◻ Buffer PE (Wash buffer), pH 7.5 ⟋ R0011
◻ 5 mM Tris-Cl, pH 8.0 (⟋ R0056 for 1 M)
◻ Spin column, silica, about 20 µg dsDNA capacity (1 / 1 rxn)

(1.) Apply up to 700 µL of supernatant to a silica spin column. Centrifuge at 11 000 × *g* for 1 min, or use a vacuum manifold.

> ***Hint:*** Silica membranes typically bind about 20 µg DNA, so it is often possible to reuse the membrane for more of the same DNA.

(2.) *Optional:* Wash the silica matrix with 450 µL Buffer PB.

> ***Hint:*** Recommended if plasmids are isolated from strains containing high endonuclease activity such as HB101 or JM109.

(3.) Wash with 650 µL Buffer PE.

(4.) Dry the silica matrix by centrifugation at 11 000 × *g* for 2 min.

(5.) Place the spin column into a clean microcentrifuge tube.

(6.) Apply 25–55 µL 5 mM Tris pH 8.0 to the membrane. Incubate for 1 min, then elute the plasmid by centrifugation at 11 000 × *g* for 1 min.

> ***Hint:*** For higher yield or large plasmids, warm elution buffer to 70 °C.

(7.) *Optional:* Repeat elution using the same or half the volume to improve yield. Pool eluates if desired.

B › *Purification by alcohol precipitation*

- ☐ Isopropyl alcohol
  - • *or*: 100% Ethanol
- ☐ 3 M NaOAc, pH 5.2 (*optional*) ◇ R0045
- ☐ Buffer PE (Wash buffer) ◇ R0011
- ☐ 5 mM Tris-Cl, pH 8.0 (◇ R0056 for 1 M)

(1.) Transfer up to 700 µL of supernatant to a clean microcentrifuge tube.

(2.) Add 0.7 vol of isopropyl alcohol, or 2 vol of ethanol, to precipitate the DNA. Vortex to mix thoroughly.

(3.) Centrifuge at 11 000 × *g* for 5 min. Discard the supernatant and briefly spin to remove residual solvent. The pellet will appear nearly transparent.

(4.) Wash with 650 µL Buffer PE or freshly prepared 70% ethanol without disturbing the pellet.

(5.) Air-dry the pellet for 2–5 min, or dry in a vacuum centrifuge for 1 min.

(6.) Resuspend in 25–55 µL 5 mM Tris pH 8.0 or water.

C › *Purification of plasmid DNA by anion-exchange chromatography*

- ☐ Buffer P3 (Neutralization buffer), pH 5.5 ◇ R0008
- ☐ Buffer QBT (Equilibration buffer), pH 7.0 ◇ R0197
- ☐ Buffer QC (Wash buffer), pH 7.0 ◇ R0198
- ☐ Buffer QF (Elution buffer), pH 8.5 ◇ R0199
- ☐ Anion-exchange column, silica-based, hydrophilic resin (1 / 1 rxn)

(1.) Lyse as in the shared procedure, but with equal volumes of Buffers P1, P2 and P3, substituting chilled Buffer P3 for Buffer N3, and incubate the neutralized lysate on ice. Take the culture and buffer volumes from the handbook of the tip in hand.

> ***Critical:*** Clear the lysate at 20 000 × *g* for 30 min at 4 °C, then spin the supernatant again for 15 min. The brief spin of the shared procedure does not clear a lysate of this scale, and carried-over precipitate clogs the column.

(2.) Equilibrate the anion-exchange column with the recommended volume of Buffer QBT and allow it to drain by gravity flow.

(3.) Apply the P3-neutralized, cleared lysate to the equilibrated column and let it pass through by gravity.

> ***Critical:*** Load only a fully cleared lysate. Carried-over precipitate clogs the resin, and the flow then stops for good; it cannot be restarted by adding buffer.

> ***This is why:*** The tip does not run dry between steps: gravity flow stops when the meniscus reaches the upper frit.

(4.) Wash the column twice with the recommended volume of Buffer QC.

(5.) Elute the plasmid DNA with the recommended volume of Buffer QF into a clean tube.

(6.) Precipitate the DNA by adding 0.7 vol isopropanol at room temperature, wash, and resuspend as described in *Alternative B*.

Analyses

- ○ Measure DNA concentration and purity by UV absorbance at 260 nm, 280 nm, and 230 nm.

Nucleic acid	$A_{260} = 1.0$	A_{260}/A_{280}	A_{260}/A_{230}
dsDNA	50 ng/µL	1.8–1.9	2.0–2.2

> ***Quality assurance:*** Extinction coefficients vary with nucleotide composition. A_{260}/A_{280} ratios increase with pH. RNA contamination elevates the ratio; protein or phenol contamination depresses it.

- ○ Confirm plasmid identity by restriction digestion SOP 0033 and agarose gel electrophoresis SOP 0032.

> ***Hint:*** Use one or more enzymes predicted to linearize the plasmid or release a diagnostic fragment.

Troubleshooting

Alkaline lysis

In Step 4:

– Chromosomal DNA does not form a white precipitate but remains viscous.
  - Reduce culture volume and repeat the preparation with gentler handling.

In Step 6:

– Chromosomal DNA in the eluate
  - Use cultures within 12–16 h post-inoculation. Older cultures release degraded chromosomal DNA. If not processing immediately, store the bacterial pellet at −20 °C.
  - Mix lysate gently after adding Buffer P2. Limit lysis to 5 min to avoid DNA shearing.
  - After adding Buffer N3, mix immediately and gently to allow complete neutralization.

– RNA contamination in eluate
  - Verify RNase A was added to Buffer P1 before use. Supplement fresh RNase if needed.

Purification of plasmid DNA over a silica matrix

In Step 6:

– Little or no DNA in eluate
  - Ensure elution buffer is low salt and within pH 7.0 and pH 8.5.

Purification by alcohol precipitation

In Step 3:

– No visible pellet after isopropanol precipitation
  - Centrifuge for longer (10 min). Mark the tube orientation before spinning to locate it on the expected side.
  - Add 0.1 vol of 3 M sodium acetate pH 5.2 before adding isopropanol to improve precipitation of dilute samples.

Purification of plasmid DNA by anion-exchange chromatography

In Step 5:

– Low yield after elution
  - Pre-warm Buffer QF to 65 °C for very large constructs (above 45–50 kbp) to improve recovery.

Materials

Alkaline lysis

- ☐ **LB medium** (**Miller**) ⊘ R0030
- ☐ **Buffer P1** (**Resuspension buffer**), pH 8.0 ⊘ R0006
  Store at 4 °C.
- ☐ **Buffer P2** (**Lysis buffer**) ⊘ R0007
  Store at room temperature.
- ☐ **Buffer N3** (**Neutralization buffer**), pH 4.8 ⊘ R0009
  Store at room temperature.

Purification of plasmid DNA over a silica matrix

- ☐ **Buffer PB** (**Binding buffer**), pH 5.0 ⊘ R0010
- ☐ **Buffer PE** (**Wash buffer**), pH 7.5 ⊘ R0011
- ☐ **5 mM Tris hydrochloride** (**Tris-Cl**), pH 8.0 (⊘ R0056 for 1 M)
- ☐ **Spin column** (1 / 1 rxn), silica, about 20 µg dsDNA capacity
  - **QIAprep spin column**
    QIAGEN, QIAprep Spin Miniprep Kit, 27104

Purification by alcohol precipitation

- ☐ **Precipitating alcohol**
  - **Isopropyl alcohol**
    CAS 67-63-0
  - *or* **Ethanol**, 100%
    CAS 64-17-5
- ☐ **Sodium acetate** (**NaOAc**), 3 M, pH 5.2 ⊘ R0045 (*optional*)
- ☐ **Buffer PE** (**Wash buffer**) ⊘ R0011
- ☐ **5 mM Tris hydrochloride** (**Tris-Cl**), pH 8.0 (⊘ R0056 for 1 M)

Purification of plasmid DNA by anion-exchange chromatography

- ☐ **Buffer P3** (**Neutralization buffer**), pH 5.5 ⊘ R0008
  Store at 4 °C.
- ☐ **Buffer QBT** (**Equilibration buffer**), pH 7.0 ⊘ R0197
  Store at room temperature.
- ☐ **Buffer QC** (**Wash buffer**), pH 7.0 ⊘ R0198
  Store at room temperature.
- ☐ **Buffer QF** (**Elution buffer**), pH 8.5 ⊘ R0199
  Store at room temperature.
- ☐ **Anion-exchange column** (1 / 1 rxn), silica-based, hydrophilic resin
  Hint: Choose a gravity-flow tip with its capacity scaled to the plasmid yield wanted (100 µg, 500 µg, 2 500 µg).
  - **QIAGEN-tip**
    QIAGEN, QIAGEN Plasmid Midi/Maxi/Mega Kit

Recipes

Buffer P1 (Resuspension buffer), pH 8.0

Amount	Ingredient		Stock	Final
12.5 mL	Tris hydrochloride (Tris-Cl), pH 8.0 R0056		1 M	50 mM
5.0 mL	EDTA, pH 8.0 R0017		0.5 M	10 mM
250 µL	RNase A, bovine pancreas R0040	[9001-99-4]	7 U/µL	7 U/mL
	Respiratory sensitization (H334)			
To 250 mL	Water, reagent-grade			

Store at 4 °C. ***Critical:*** If the buffer is older than six months, supplement with fresh RNase A.

Buffer P1 (Resuspension buffer)

50 mM Tris-Cl, 10 mM EDTA, 7 U/mL RNase A, pH 8.0

No GHS hazards at the indicated concentrations

☐ Hold for EHS review before disposal

Date: Sign: R0006

Buffer P2 (Lysis buffer)

Amount	Ingredient		Stock	Final
2.0 g	NaOH	[1310-73-2]	40.00 g/mol	200 mM
	Skin corrosion (H314)			
12.5 mL	SDS R0047		20%	1%
	Skin irritation (H315); Serious eye damage (H318)			
125 mg	Thymolphthalein	☐ [125-20-2]	430.54 g/mol	0.05%
To 250 mL	Water, reagent-grade			

Dissolve the sodium hydroxide first and add the SDS last. Store at room temperature. ***Critical:*** Add the SDS last and do not shake the solution; sodium hydroxide dissolved in an SDS solution foams, and the detergent precipitates in the cold. ***This is why:*** Thymolphthalein serves as pH indicator to monitor the neutralization reaction with N3; it is optional.

Buffer P2 (Lysis buffer)

200 mM NaOH, 1% SDS, ☐ 0.05% Thymolphthalein

WARNING

Skin and serious eye irritation

☐ Collect as HAZARDOUS WASTE
☐ DO NOT wash into sewer

Date: Sign: R0007

Buffer N3 (Neutralization buffer), pH 4.8

Amount	Ingredient		Stock	Final
100.0 g	Guanidine · HCl	[50-01-1]	95.53 g/mol	4.2 M
22.0 g	KOAc	[127-08-2]	98.14 g/mol	0.9 M
To 250 mL	Water, reagent-grade			

Adjust pH with acetic acid. Store at room temperature. ***Hint:*** For use with silica-gel membranes as in the QIAGEN QIAprep Spin Miniprep Kit.

Buffer N3 (Neutralization buffer)

4.2 M Guanidine, 0.9 M KOAc, pH 4.8

WARNING

Skin and serious eye irritation; Acute oral toxicity

☐ Neutralise to pH 6–9
☐ Flush to drain with excess water

Date: Sign: R0009

Buffer PB (Binding buffer), pH 5.0

Amount	Ingredient		Stock	Final
119.0 g	Guanidine · HCl	[50-01-1]	95.53 g/mol	5.0 M
75 mL	Isopropyl alcohol	[67-63-0]	100%	30%
	Flammable liquid (H225)			
To 250 mL	Water, reagent-grade			

Adjust pH with acetic acid before adding isopropyl alcohol. Store at room temperature.

Buffer PB (Binding buffer)

5.0 M Guanidine, 30% Isopropyl alcohol, pH 5.0



DANGER

Flammable liquid; Skin and serious eye irritation; Drowsiness or dizziness; Acute oral toxicity

- ☐ Collect as ORGANIC SOLVENT WASTE
- ☐ Keep away from oxidizers

Date: Sign: R0010

Buffer PE (Wash buffer), pH 7.5

Amount	Ingredient		Stock	Final
2.5 mL	Tris-Cl, pH 7.5	R0055	1 M	10 mM
200 mL	Ethanol, reagent-grade	[64-17-5]	100%	80%
	Flammable liquid (H225)			
To 250 mL	Water, reagent-grade			

Store at room temperature.

Buffer PE (Wash buffer)

10 mM Tris-Cl, 80% Ethanol, pH 7.5



DANGER

Flammable liquid

- ☐ Collect as ORGANIC SOLVENT WASTE
- ☐ Keep away from oxidizers

Date: Sign: R0011

Buffer P3 (Neutralization buffer), pH 5.5

Amount	Ingredient		Stock	Final
73.6 g	KOAc	[127-08-2]	98.14 g/mol	3.0 M
To 250 mL	Water, reagent-grade			

Adjust pH with acetic acid. Store at 4 °C. ***Hint:*** For use with anion-exchange columns as in the QIAGEN Plasmid Kit.

Buffer P3 (Neutralization buffer)

pH 5.5





EPA Safer Choice low-concern

- ☐ Flush to drain with excess water

Date: Sign: R0008

Buffer QBT (Equilibration buffer), pH 7.0

Amount	Ingredient		Stock	Final
10.95 g	NaCl	[7647-14-5]	58.44 g/mol	750 mM
2.62 g	MOPS	[1132-61-2]	209.26 g/mol	50 mM
37.5 mL	Isopropyl alcohol	[67-63-0]	100%	15%
	Flammable liquid (H225)			
375 µL	Triton™ X-100	[9036-19-5]	100%	0.15%
	Serious eye damage (H318)			
To 250 mL	Water, reagent-grade			

Adjust pH with sodium hydroxide before adding isopropanol. Store at room temperature.

Buffer QBT (Equilibration buffer)

750 mM NaCl, 50 mM MOPS, 15% Isopropyl alcohol, 0.15% Triton™ X-100, pH 7.0



DANGER

Flammable liquid; Serious eye irritation; Drowsiness or dizziness

- ☐ Collect as ORGANIC SOLVENT WASTE
- ☐ Keep away from oxidizers

Date: Sign: R0197

Buffer QC (Wash buffer), pH 7.0

Amount	Ingredient		Stock	Final
2.62 g	MOPS	[1132-61-2]	209.26 g/mol	50 mM
14.61 g	NaCl	[7647-14-5]	58.44 g/mol	1.0 M
37.5 mL	Isopropyl alcohol	[67-63-0]	100%	15%
	Flammable liquid (H225)			
To 250 mL	Water, reagent-grade			

Adjust pH with sodium hydroxide before adding isopropanol. Store at room temperature.

Buffer QC (Wash buffer)

50 mM MOPS, 1.0 M NaCl, 15% Isopropyl alcohol, pH 7.0



DANGER

Flammable liquid; Serious eye irritation; Drowsiness or dizziness

- ☐ Collect as ORGANIC SOLVENT WASTE
- ☐ Keep away from oxidizers

Date: Sign: R0198

Buffer QF (Elution buffer), pH 8.5

Amount	Ingredient		Stock	Final
1.51 g	Tris base	[77-86-1]	121.14 g/mol	50 mM
18.26 g	NaCl	[7647-14-5]	58.44 g/mol	1.25 M
37.5 mL	Isopropyl alcohol	[67-63-0]	100%	15%
	Flammable liquid (H225)			
To 250 mL	Water, reagent-grade			

Adjust to pH 8.5 with hydrochloric acid before adding isopropanol. Store at room temperature.

Buffer QF (Elution buffer)

50 mM Tris base, 1.25 M NaCl, 15% Isopropyl alcohol, pH 8.5



DANGER

Flammable liquid; Serious eye irritation; Drowsiness or dizziness

- ☐ Collect as ORGANIC SOLVENT WASTE
- ☐ Keep away from oxidizers

Date: Sign: R0199

List of references

H. Birnboim and J. Doly, *Nucleic Acids Res.* **7**(6), 1513—1523 (1979).

Change log

2016-03-15 **Rahul Patharkar** Alkaline lysis with ethanol precipitation; from the Patharkar life-science protocols hub (patharkar.com).

2023-05-30 **Benjamin C. Buchmuller** Adaptation as SOP, supplemented from Qiagen "QIAprep Spin Miniprep Kit" handbook.

2026-05-21 **Benjamin C. Buchmuller** Fix two transcription errors in Buffer P1: RNase A final concentration (was "7 enzymeunit per liter", now "7 enzymeunit per milliliter"; stock and amount unchanged, prep unaffected) and EDTA amount (was 10.0 mL, now 5.0 mL — recipe v1 over-prepared at 20 mM; corrected recipe yields the intended 10 mM, matching the Qiagen reference).

2026-10-26 **Benjamin C. Buchmuller** Corrected the anion-exchange procedure, Buffer PB to pH 5.0 and the run-dry caution; added the lysis conditions procedure C requires and the guanidine and bleach hazard. The per-tip culture, buffer and column volumes are withdrawn: they belong to whichever midi, maxi or mega product is bought, and those are far less standardized than the mini columns. Take them from the handbook of the tip in hand. Credits: QIAGEN for Buffers P1, P2, P3, QBT, QC and QF; the Daniel Summerer lab and Tom Muir lab sheets for N3, PB and PE.

From the *Lab Protocols* collection by B. C. Buchmuller and contributors (2026). Licensed under Creative Commons Attribution Share Alike 4.0 International; see https://creativecommons.org/licenses/by-sa/4.0/ for the terms.

For research use. Provided in good faith, without warranty or liability for any use or results. Users are responsible for compliance with local regulations and institutional policies. Where a brand name, manufacturer, or specific source is cited, it is given as an example and does not imply that other suitable equivalents are excluded.

Current when printed. Visit https://benjbuch.github.io/check/ or scan the QR code to check for updates.

0a925f9